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(54) **STORAGE DEVICE AND METHOD PERFORMED BY THE SAME**

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(57) **ABSTRACT**

A storage device and a method performed by the same are provided in the present disclosure, the storage device including: a memory comprising a plurality of virtual function (VF) apparatuses; a controller configured to: initialize a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and control reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and data size of the I/O command.

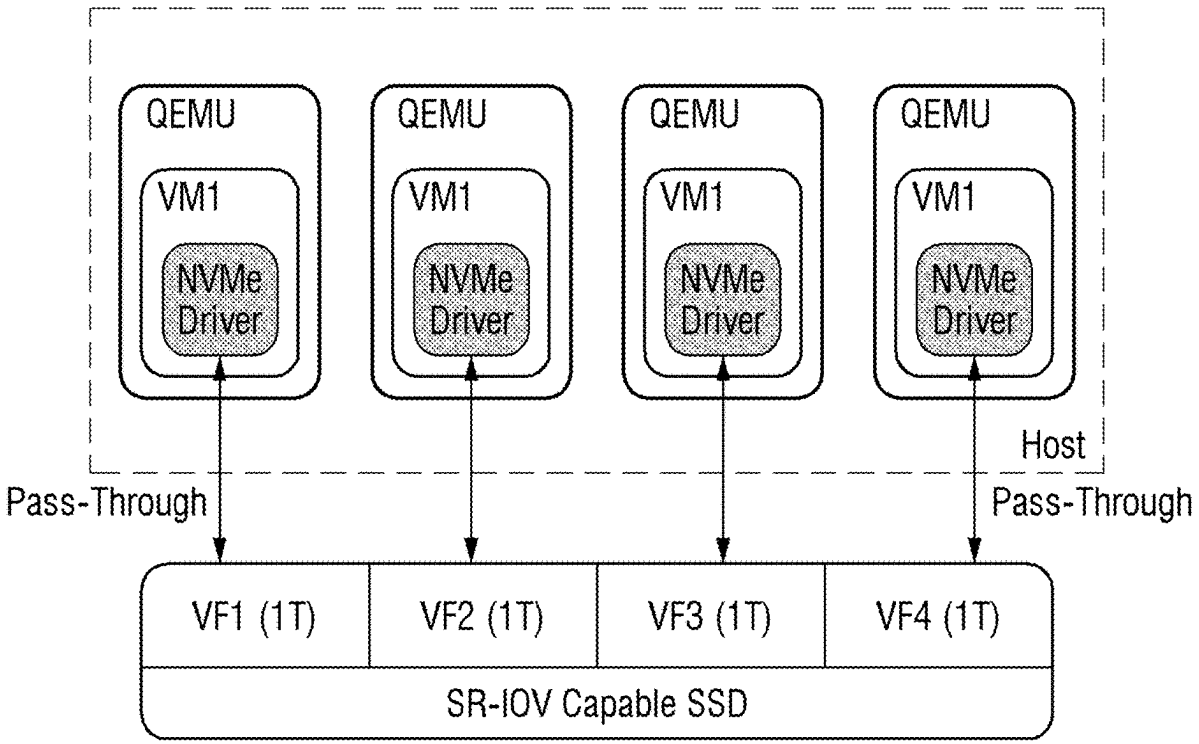


Fig. 1

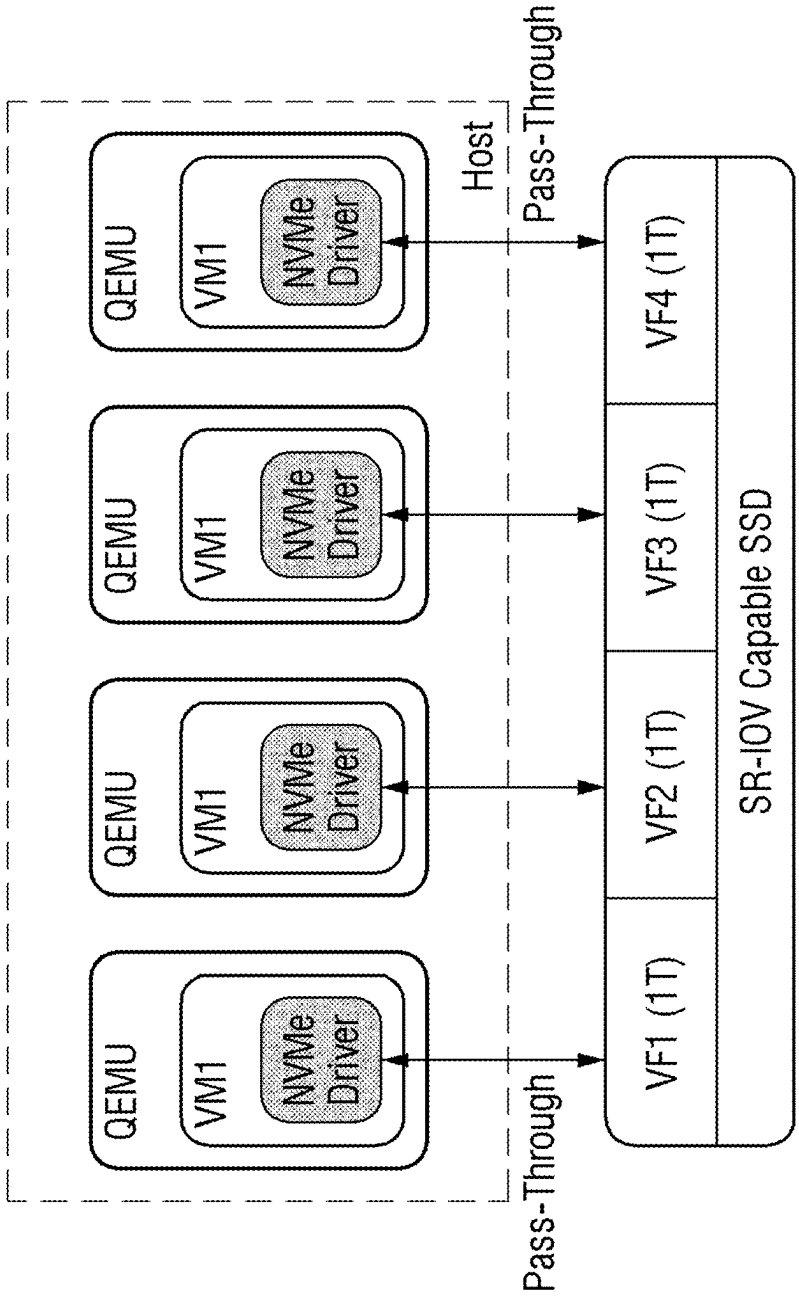

QEMU
VM1
NVMe Driver
QEMU
VM1
NVMe Driver
QEMU
VM1
NVMe Driver
QEMU
VM1
NVMe Driver
Host
Pass-Through
Pass-Through
VF1 (1T)
VF2 (1T)
VF3 (1T)
VF4 (1T)
SR-IOV Capable SSD

Fig. 2

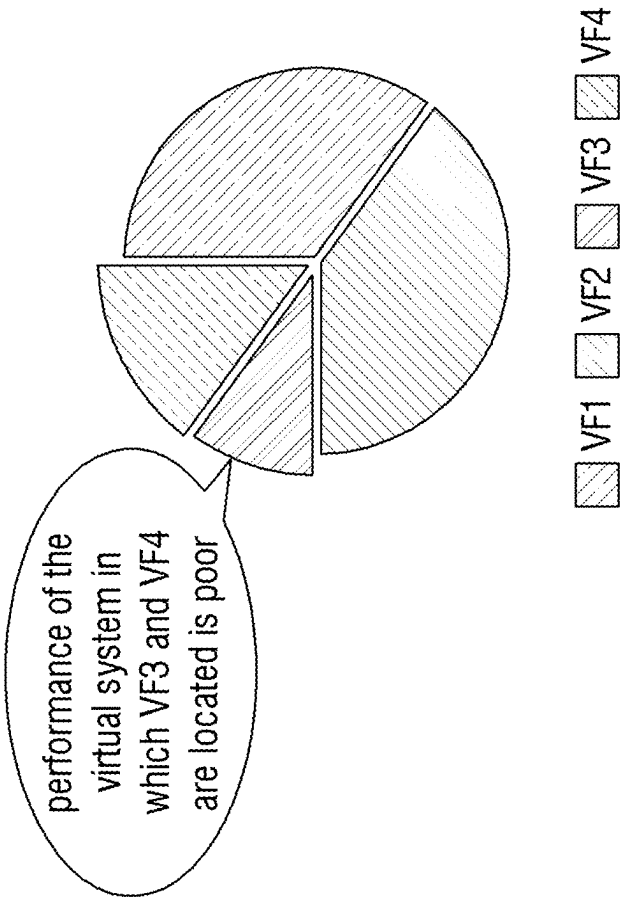

load :
VF1: 128k sequential read
VF2: 128k sequential write
VF3: 4k random read
VF4: 4k random write
VF1
VF2
VF3
VF4
performance of the virtual system in which VF3 and VF4 are located is poor

Fig. 3

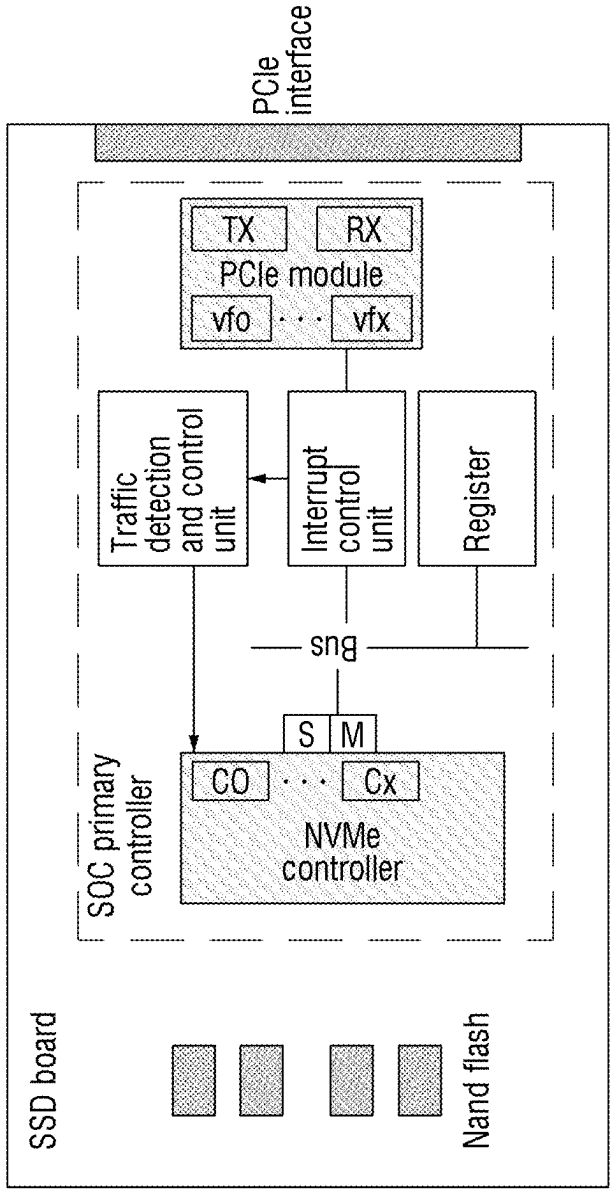

PCIe interface
TX
RX
PCIe module
vf0
vfx
Traffic detection and control unit
Interrupt control unit
Register
sng
S
M
SOC primary controller
C0
Cx
NVMe controller
SSD board
Nand flash

Fig. 4

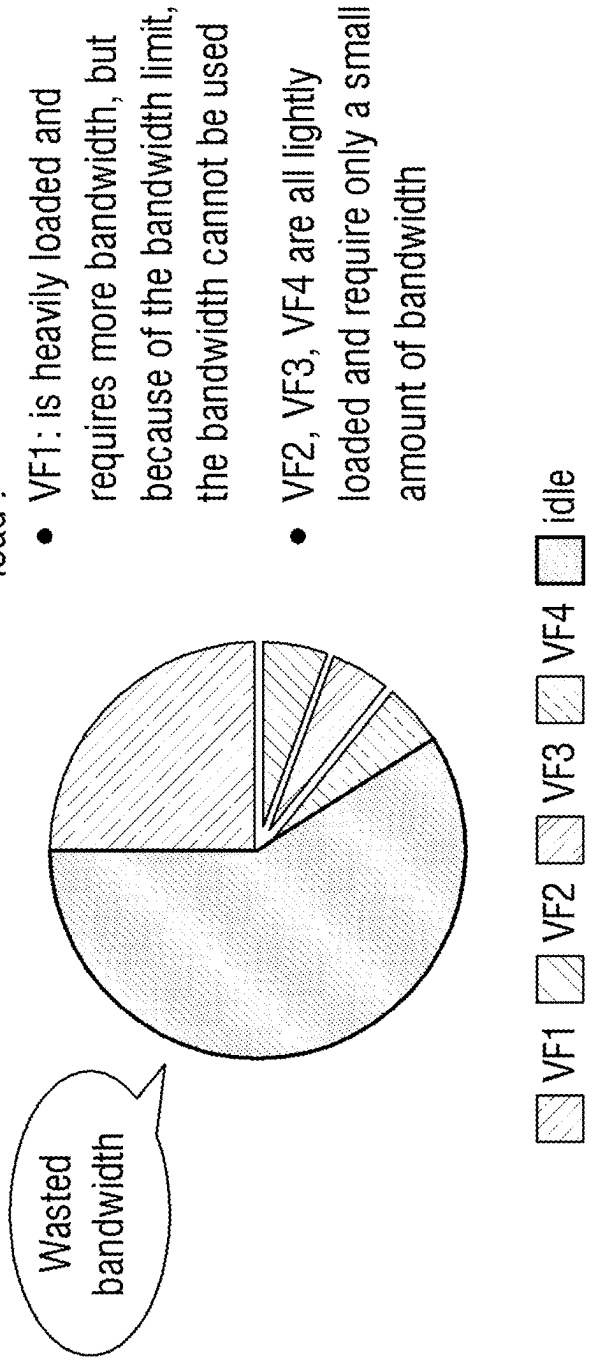

load :
• VF1: is heavily loaded and requires more bandwidth, but because of the bandwidth limit, the bandwidth cannot be used
• VF2, VF3, VF4 are all lightly loaded and require only a small amount of bandwidth
Wasted bandwidth
VF1
VF2
VF3
VF4
idle

Fig. 5

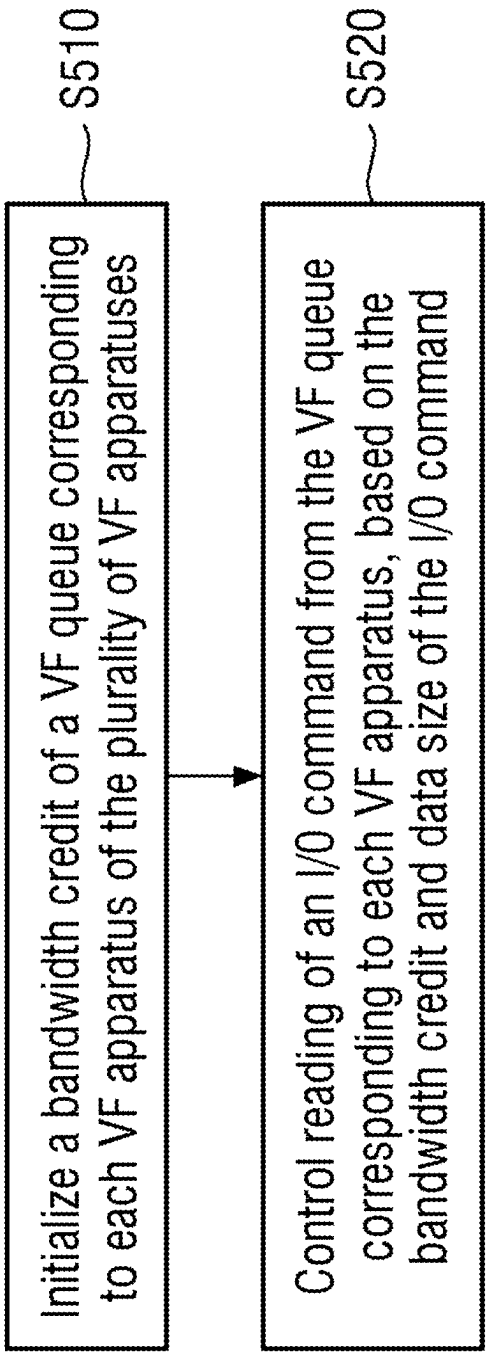

S510
Initialize a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses
S520
Control reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and data size of the I/O command

Fig. 6

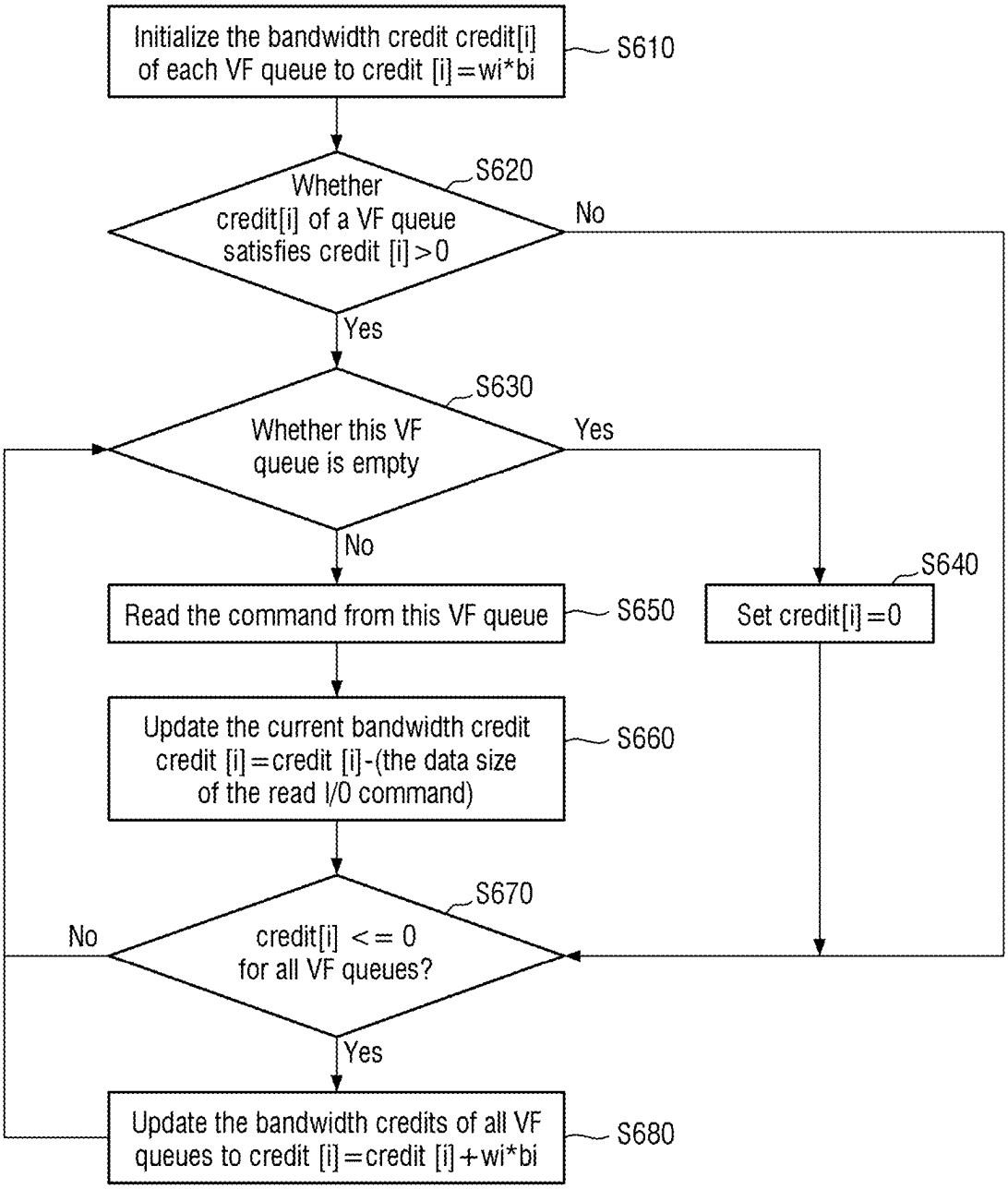
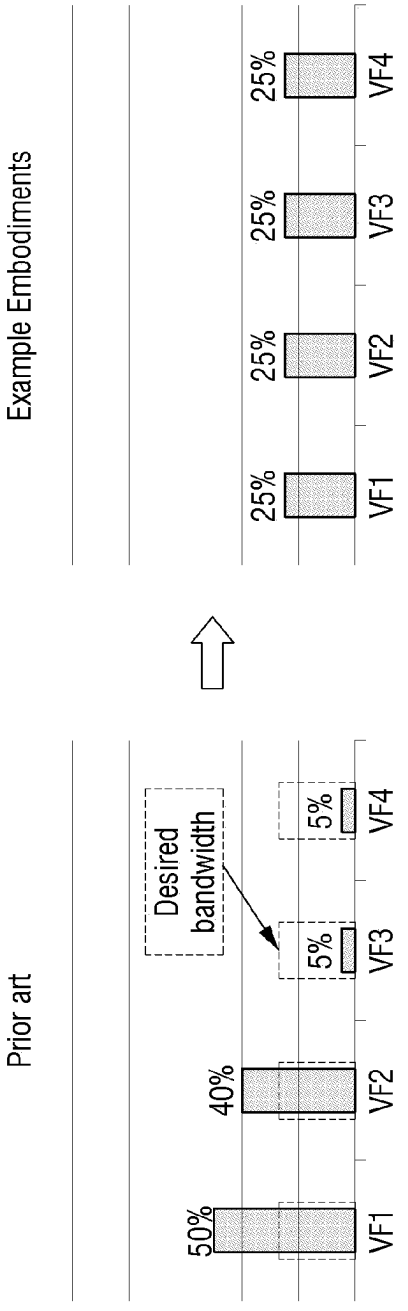

Initialize the bandwidth credit credit[i] of each VF queue to credit [i]=wi*bi
S610
Whether credit[i] of a VF queue satisfies credit [i]>0
S620
No
Yes
Whether this VF queue is empty
S630
Yes
No
Read the command from this VF queue
S650
Set credit[i]=0
S640
Update the current bandwidth credit credit [i]=credit [i]-(the data size of the read I/O command)
S660
credit[i] <= 0 for all VF queues?
S670
No
Yes
Update the bandwidth credits of all VF queues to credit [i]=credit [i]+wi*bi
S680

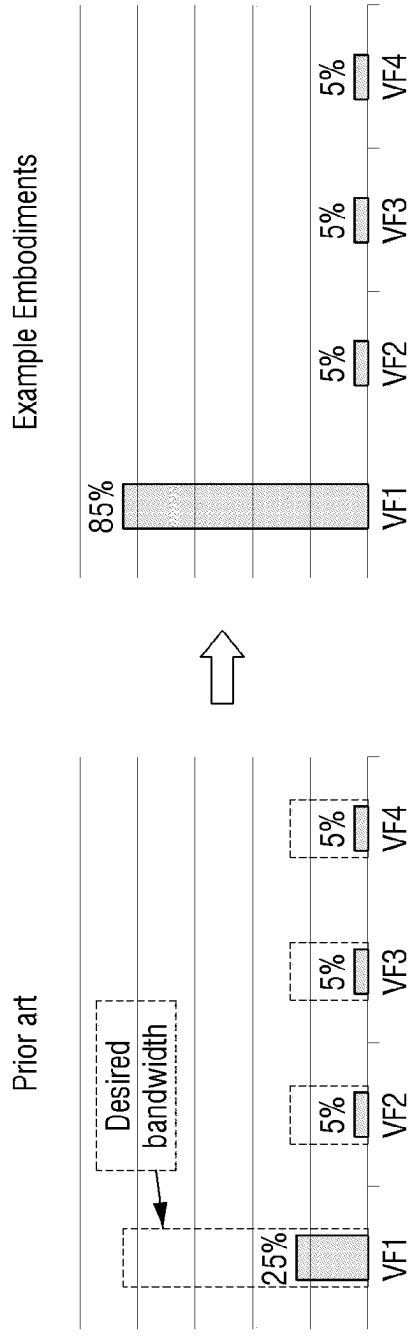
Fig. 7



	Desired value	actual value
VF1	25%	50%
VF2	25%	40%
VF3	25%	5%
VF4	25%	5%

	Desired value	actual value
VF1	25%	25%
VF2	25%	25%
VF3	25%	25%
VF4	25%	25%

Fig. 8



	Desired value	actual value
VF1	85%	25%
VF2	5%	5%
VF3	5%	5%
VF4	5%	5%

	Desired value	actual value
VF1	85%	85%
VF2	5%	5%
VF3	5%	5%
VF4	5%	5%

Fig. 9

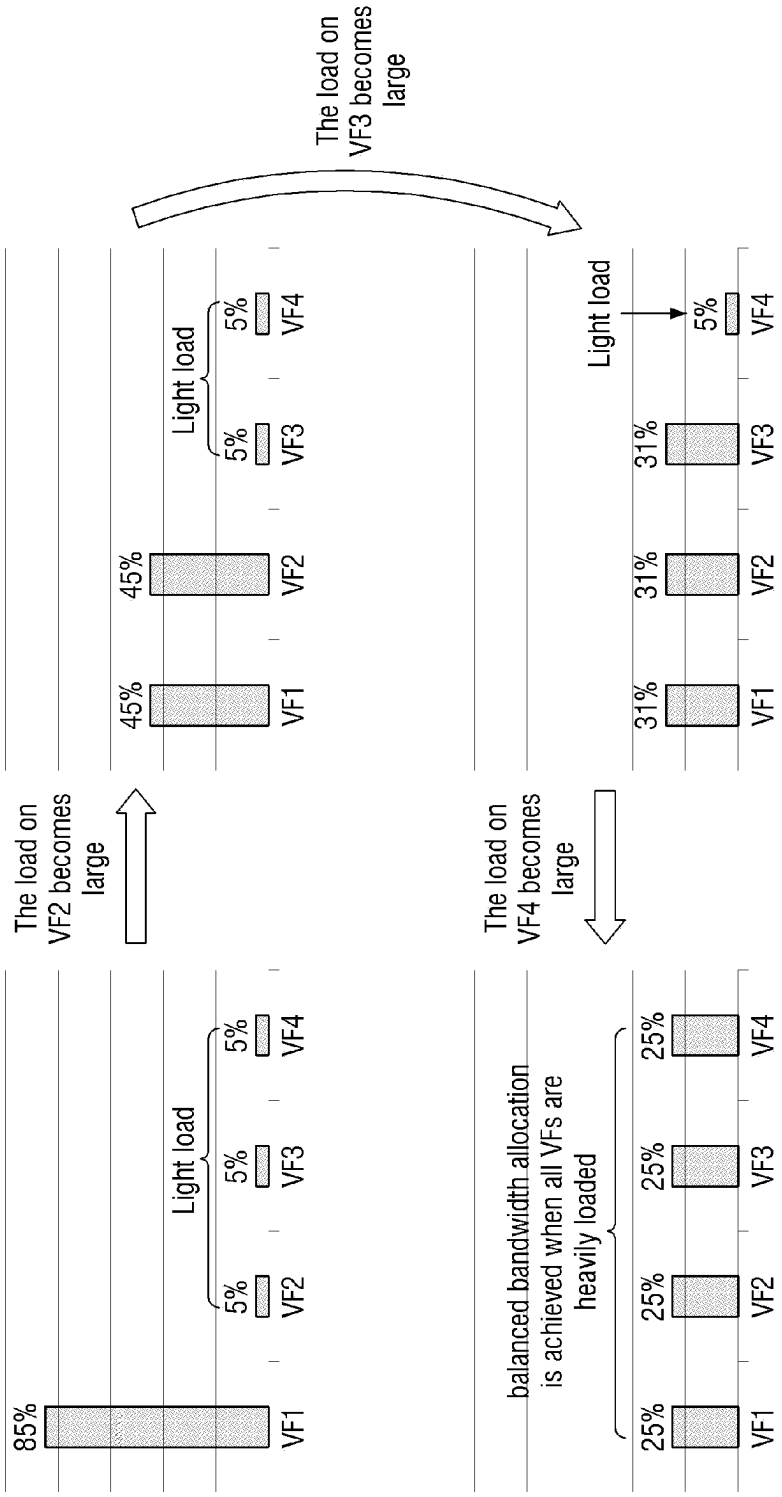

85%
5%
5%
Light load
VF1
VF2
VF3
VF4
The load on VF2 becomes large
45%
45%
5%
5%
Light load
VF1
VF2
VF3
VF4
The load on VF3 becomes large
31%
31%
31%
5%
Light load
VF1
VF2
VF3
VF4
The load on VF4 becomes large
balanced bandwidth allocation is achieved when all VFs are heavily loaded
25%
25%
25%
25%
VF1
VF2
VF3
VF4

Fig. 10

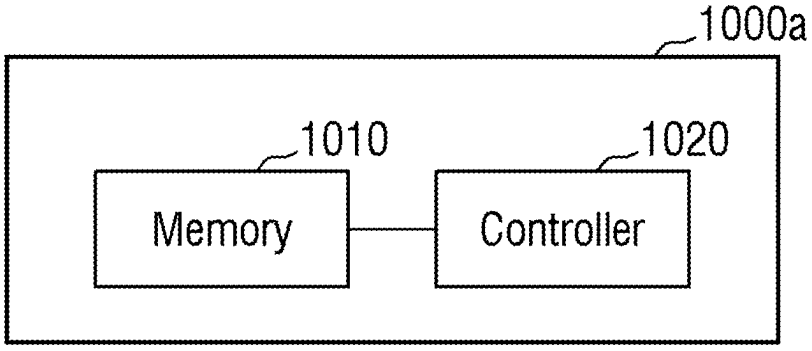

1000a
1010
1020
Memory
Controller

Fig. 11

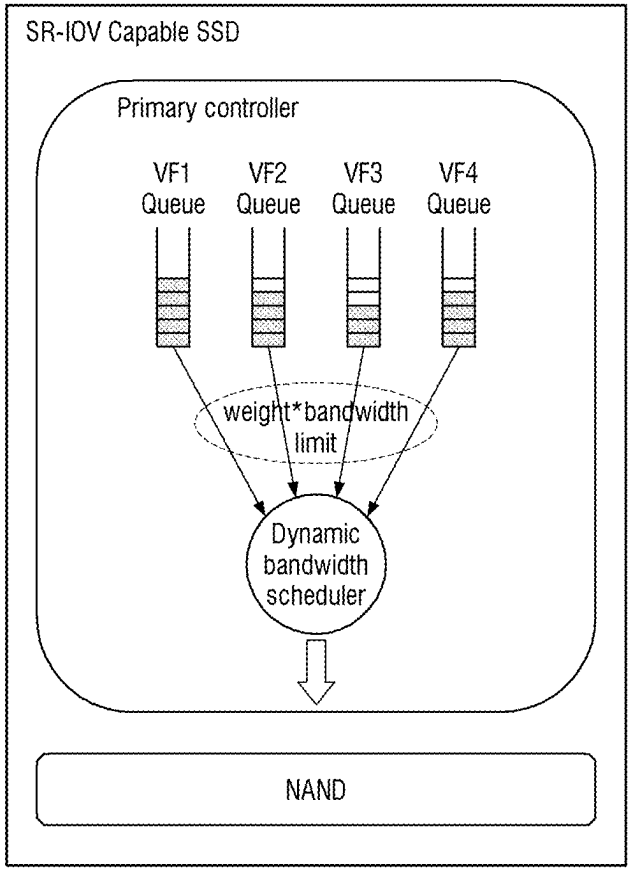

SR-IOV Capable SSD
Primary controller
VF1 Queue
VF2 Queue
VF3 Queue
VF4 Queue
weight*bandwidth limit
Dynamic bandwidth scheduler
NAND

Fig. 12

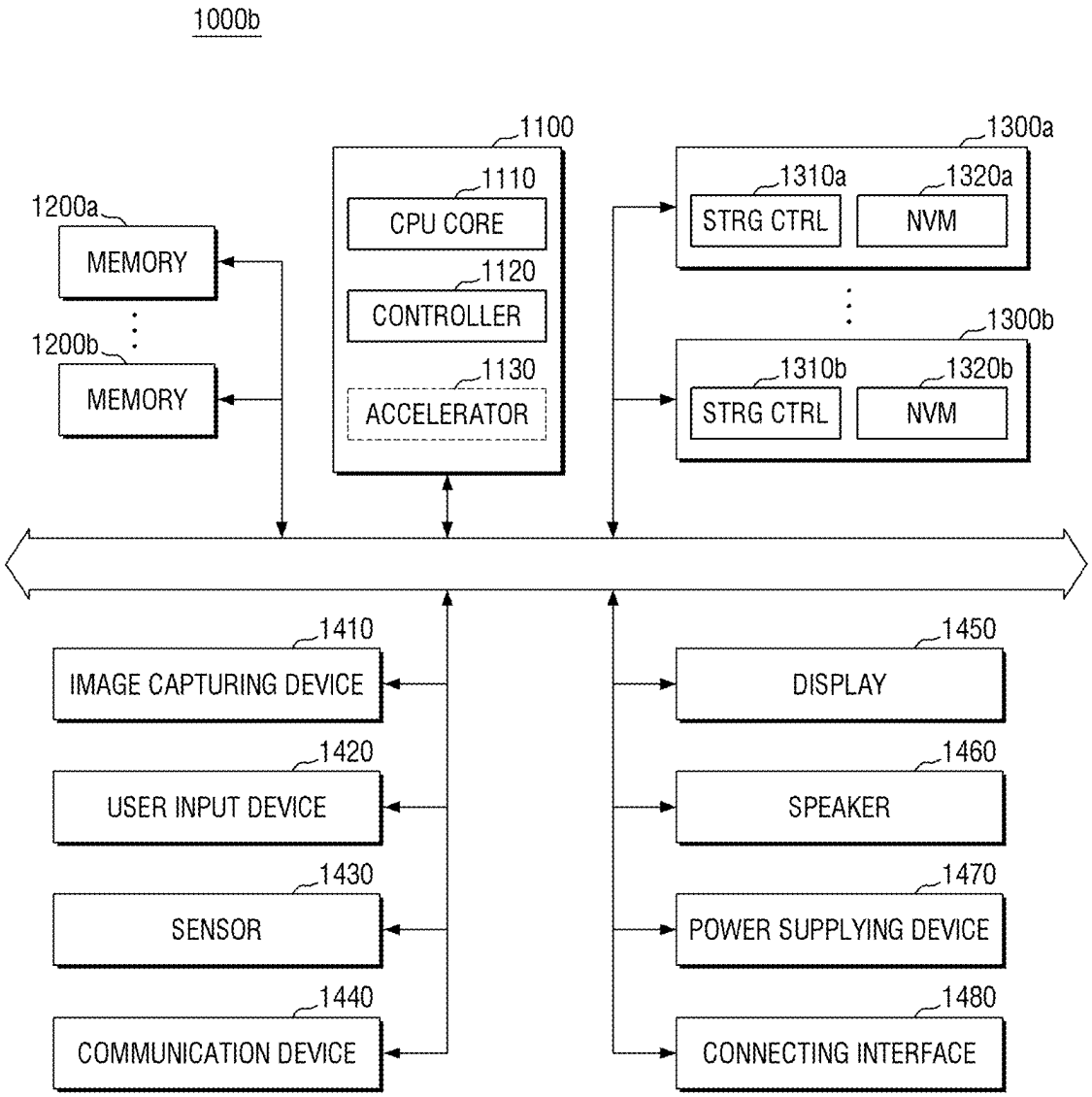

1000b
1100
1110
CPU CORE
1120
CONTROLLER
1130
ACCELERATOR
1200a
MEMORY
1200b
MEMORY
1300a
1310a
STRG CTRL
1320a
NVM
1300b
1310b
STRG CTRL
1320b
NVM
1410
IMAGE CAPTURING DEVICE
1420
USER INPUT DEVICE
1430
SENSOR
1440
COMMUNICATION DEVICE
1450
DISPLAY
1460
SPEAKER
1470
POWER SUPPLYING DEVICE
1480
CONNECTING INTERFACE

Fig. 13

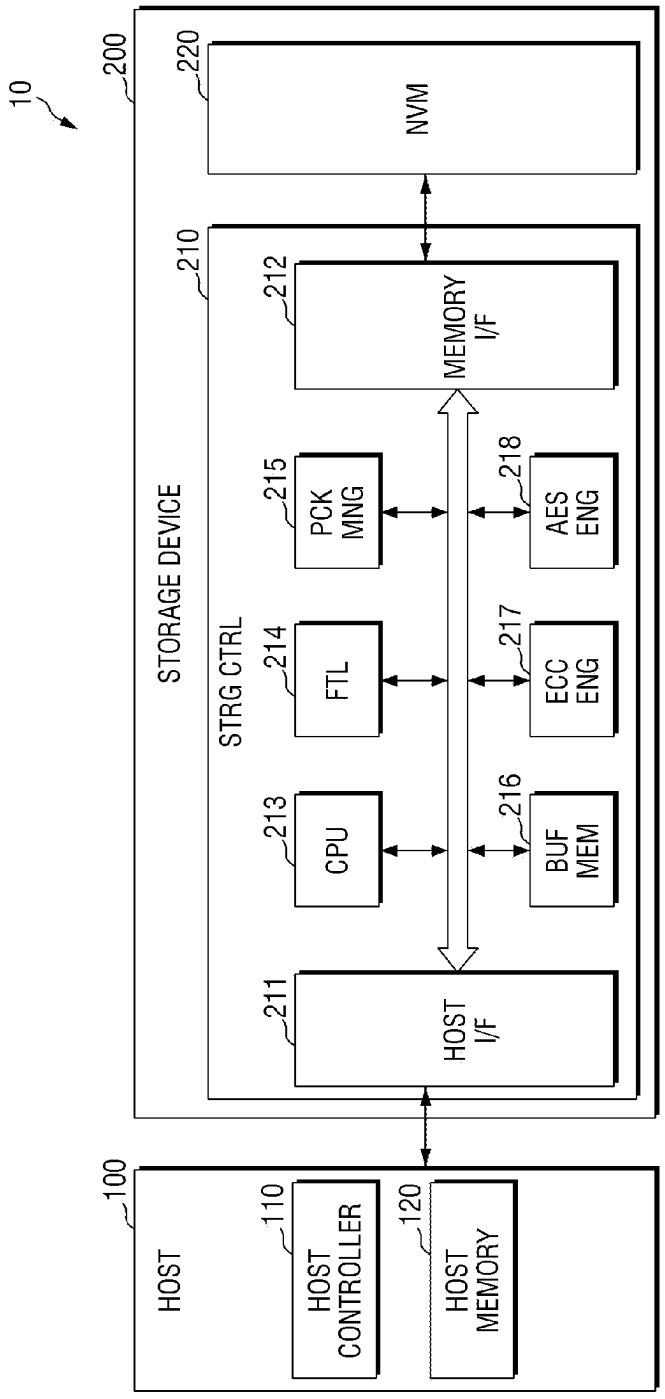
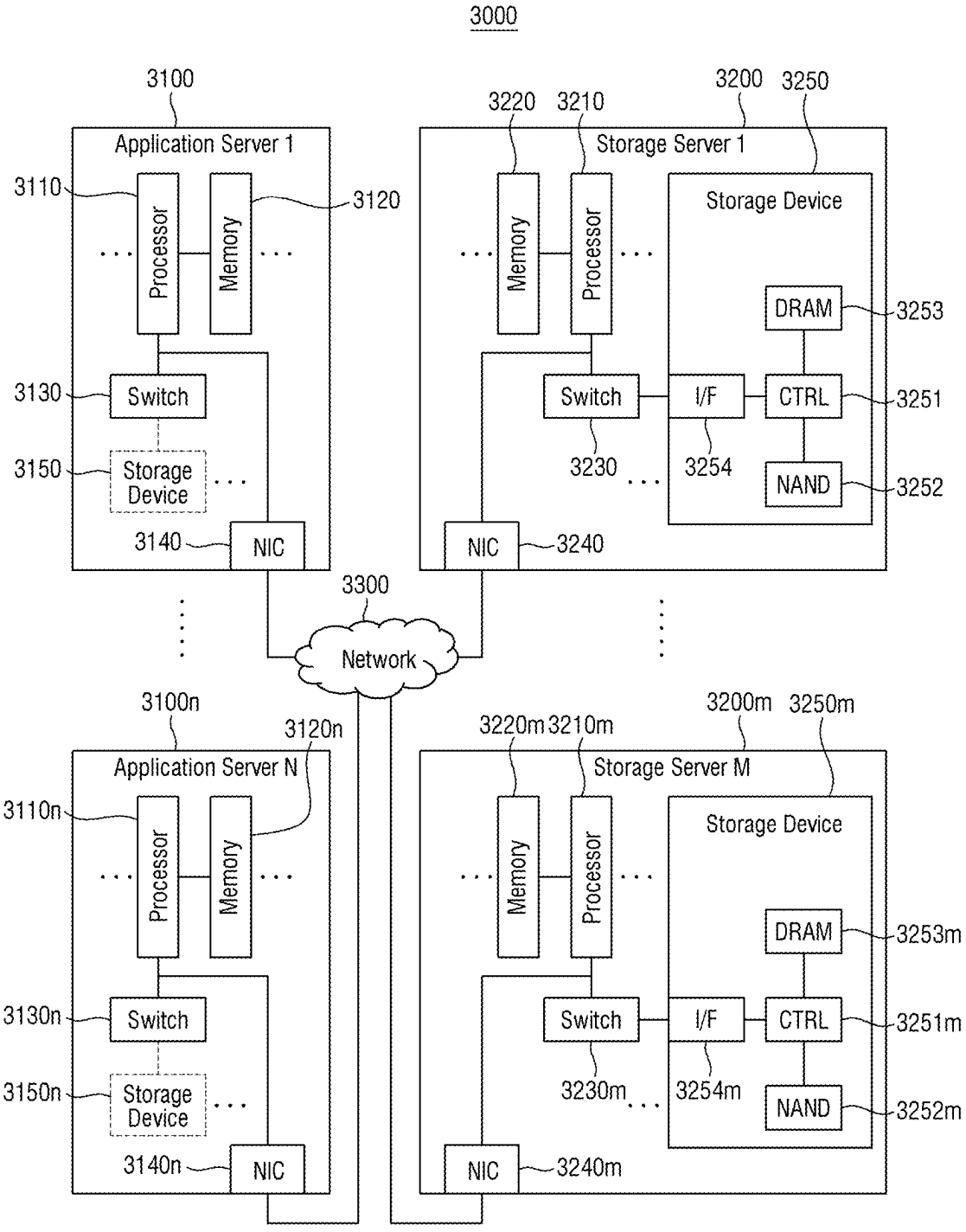

10
200
220
NVM
210
212
MEMORY I/F
STORAGE DEVICE
STRG CTRL
215
PCK MNG
218
AES ENG
214
FTL
217
ECC ENG
213
CPU
216
BUF MEM
211
HOST I/F
100
HOST
110
HOST CONTROLLER
120
HOST MEMORY

Fig. 14


3000
3100
Application Server 1
3110
Processor
Memory
3120
3130
Switch
3150
Storage Device
3140
NIC
3200
3250
3220
3210
Storage Server 1
Storage Device
DRAM
3253
Switch
I/F
CTRL
3251
3230
3254
NAND
3252
NIC
3240
3300
Network
3100n
Application Server N
3120n
3110n
3130n
3150n
3140n
3200m
3250m
3220m
3210m
Storage Server M
3253m
3251m
3230m
3254m
3252m
3240m

STORAGE DEVICE AND METHOD PERFORMED BY THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Chinese Patent Application No. 202410276885.1, filed on Mar. 11, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of semiconductors and, more specifically, to a storage device, a method performed by the storage device, a storage system, a host storage system, a data center system, a computer readable storage medium and an electronic apparatus.

BACKGROUND

[0003] In an SR-IOV (Single Root I/O Virtualization) based I/O virtualization architecture, a plurality of apparatuses may share a single storage device, for example, an SSD following the PCIe protocol. This physical PCIe apparatus (Physical Function, PF) may present a plurality of Virtual Function (Virtual Function, VF) apparatuses to a host, whereby a single PCIe storage apparatus may include a plurality of VF apparatuses, each of which may be connected directly to a virtual machine of the host. Although this approach may reduce the involvement of a virtual machine monitor in I/O operations and improve I/O performance of the virtual machine, in such an SR-IOV-based architecture, when the load of each VF apparatus is large, there may be competition for bandwidth among the VF apparatuses, and bandwidth resources will then be unevenly distributed, with some VF apparatuses grabbing more bandwidth resources which causes other VF apparatuses to occupy very little bandwidth. For these VF apparatuses with little bandwidth resources, quality of service cannot be guaranteed and user experience may be poor. However, an existing attempt to solve the uneven bandwidth distribution requires the addition of new hardware for detecting changes in bandwidth traffic and a speed limit for each VF apparatus. Although it is possible to ensure that each VF apparatus average bandwidth in the case of bandwidth competition, there is an increase in hardware costs and/or a potential waste of bandwidth resources.

SUMMARY

[0004] According to an aspect of the present disclosure, there is provided a storage device, the storage device includes: a memory including a plurality of virtual function (VF) apparatuses; a controller configured to: initialize a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and control reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and data size of the I/O command.

[0005] Alternatively, the controller is configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on a capacity ratio of each VF apparatus occupying the memory.

[0006] Alternatively, the controller is configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on the capacity ratio and a weight value (for example, predefined) for each VF apparatus.

[0007] Alternatively, the controller is configured to: poll each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and in response to the I/O command being read out of the non-empty VF queue, update the current bandwidth credit of this VF queue by subtracting the data size of the read command from the current bandwidth credit of this VF queue, and in response to the updated current bandwidth credit of this VF queue being no longer greater than the threshold, stop reading of the I/O command from this VF queue.

[0008] Alternatively, the controller is further configured to: in response to each VF queue being polled once, determine whether current bandwidth credits of all VF queues are all less than or equal to the threshold; in response to the current bandwidth credits of all the VF queues all being less than or equal to the threshold, initialize the bandwidth credit of each VF queue again, otherwise continue to read the I/O command from the non-empty VF queue with the current bandwidth credit greater than the threshold.

[0009] Alternatively, the controller is further configured to: set the current bandwidth credit of a VF queue to the threshold if this VF queue is empty.

[0010] Alternatively, in response to the current bandwidth credits of all the VF queues being less than or equal to the threshold, the controller is configured to initialize the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue.

[0011] According to another aspect of the present disclosure, there is provided a method performed by a storage device, wherein the storage device includes a memory, the memory including a plurality of virtual function (VF) apparatuses, the method including: initializing a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and controlling reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and the data size of the I/O command.

[0012] Alternatively, the initializing the bandwidth credit of the VF queue corresponding to each VF apparatus of the plurality of VF apparatuses includes: initializing the bandwidth credit of the VF queue corresponding to each VF apparatus based on a capacity ratio of each VF apparatus occupying the memory.

[0013] Alternatively, the initializing the bandwidth credit of the VF queues corresponding to each VF apparatus based on the capacity ratio of each VF apparatus occupying the memory includes: initializing the bandwidth credit of the VF queue corresponding to each VF apparatus based on the capacity ratio and a weight value (for example, predefined) for each VF apparatus.

[0014] Alternatively, the controlling the reading of the I/O command from the VF queue corresponding to each VF apparatus based on the bandwidth credit and the data size of the I/O commands includes: polling each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and in response to the I/O command being read out of the non-empty VF queue, updating the

current bandwidth credit of this VF queue by subtracting the data size of the read command from the current bandwidth credit of this VF queue, and in response to the updated current bandwidth credit of this VF queue being no longer greater than the threshold, stopping reading of the I/O command from that VF queue.

[0015] Alternatively, the method further including: in response to each VF queue being polled once, determine whether current bandwidth credits of all VF queues are all less than or equal to the threshold; in response to the current bandwidth credits of all the VF queues all being less than or equal to the threshold, initializing the bandwidth credit of each VF queue again, otherwise continuing to read I/O commands from non-empty VF queues with the current bandwidth credit greater than the threshold.

[0016] Alternatively, the method further including: setting the current bandwidth credit of a VF queue to the threshold if this VF queue is empty.

[0017] Alternatively, the initializing the bandwidth credit of each VF queue again includes: initializing the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue.

[0018] According to another aspect of the present disclosure, there is provided a storage system, the storage system includes: a main processor; a memory; and a storage device, wherein the storage device is configured to perform the method described above.

[0019] According to another aspect of the present disclosure, there is provided a host storage system, the host storage system includes: a host; and a storage device, wherein the storage device is configured to perform the method described above.

[0020] According to another aspect of the present disclosure, there is provided a data center system, the data center system includes: a plurality of application servers; and a plurality of storage servers, wherein each storage server includes a storage device, wherein the storage device is configured to perform the method described above.

[0021] According to another aspect of the present disclosure, there is provided a computer readable storage medium having a computer program stored thereon, wherein the method described above is implemented when the computer program is executed by a processor.

[0022] According to another aspect of the present disclosure, there is provided an electronic apparatus, the electronic apparatus includes: a processor; and a memory storing a computer program, the computer program, when executed by the processor, implementing the method described above.

[0023] According to the technical solutions provided by example embodiments of the present disclosure, dynamic bandwidth resource scheduling may be achieved, which not only enables balanced bandwidth distribution, but also avoids bandwidth wastage.

[0024] It should be understood that the above general description and the later detailed description are examples and explanatory only and do not limit the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The accompanying drawings herein are incorporated into and form part of the specification, illustrate example embodiments consistent with the present disclosure, which are used in conjunction with the specification to explain the principles of the present disclosure and do not constitute an undue limitation of the present disclosure.

[0026] FIG. **1** is a schematic diagram illustrating an SR-IOV-based I/O virtualization architecture;

[0027] FIG. **2** is a schematic diagram illustrating an SR-IOV-based I/O virtualization architecture with uneven bandwidth distribution due to I/O competition;

[0028] FIG. **3** is a schematic diagram of a device that attempts to achieve balanced bandwidth by adding hardware apparatuses;

[0029] FIG. **4** is a schematic diagram illustrating bandwidth wastage;

[0030] FIG. **5** is a flowchart illustrating a method performed by a storage device according to example embodiments of the present disclosure;

[0031] FIG. **6** is a flowchart illustrating an example of a method performed by a storage device;

[0032] FIG. **7** is a schematic diagram of applying the method according to example embodiments of the present disclosure to achieve balanced bandwidth distribution;

[0033] FIG. **8** is a schematic diagram of applying a method according to example embodiments of the present disclosure to avoid a waste of bandwidth resources;

[0034] FIG. **9** is a schematic diagram of the procedure of applying the method according to example embodiments of the present disclosure for dynamic scheduling of bandwidth resources when the load varies;

[0035] FIG. **10** illustrates a block diagram of a storage device according to example embodiments of the present disclosure;

[0036] FIG. **11** illustrates an example schematic diagram of a storage device according to example embodiments of the present disclosure;

[0037] FIG. **12** is a diagram of a system to which a storage device is applied according to example embodiments of the present disclosure;

[0038] FIG. **13** is a block diagram of a host storage system according to example embodiments;

[0039] FIG. **14** is a diagram of a data center to which a memory device is applied according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

[0040] In order to enable a person of ordinary skill in the art to better understand the technical solutions of the present disclosure, the technical solutions provide by example embodiments of the present disclosure will be clearly and completely described below in conjunction with the accompanying drawings.

[0041] It should be noted that the terms "first", "second", etc. in the specification and claims of the present disclosure and the accompanying drawings above are used to distinguish similar objects rather than to describe a particular order or sequence. It should be understood that data so distinguished may be interchanged, where appropriate, so that example embodiments of the present disclosure described herein may be implemented in an order other than those illustrated or described herein. Example embodiments described in the following examples do not represent all example embodiments that are consistent with the present disclosure. Rather, they are only examples of devices and methods that are consistent with some aspects of the present disclosure, as detailed in the appended claims.

[0042] It should be noted herein that "at least one of the several items" in this present disclosure includes "any one of the several items", "any combination of the several items" and "all of the several items" the juxtaposition of these three categories. For example, "including at least one of A and B" includes the following three juxtapositions: (1) including A; (2) including B; (3) including A and B. Another example is "performing at least one of operation one and operation two", which means the following three juxtapositions (1) performing operation one; (2) performing operation two; (3) performing operation one and operation two.

[0043] For example, an SR-IOV (Single Root I/O Virtualization) based I/O virtualization architecture may have a case of uneven distribution of bandwidth resources. FIG. **1** is a schematic diagram illustrating an SR-IOV-based I/O virtualization architecture. As shown in FIG. **1**, a host includes a plurality of virtual machines (VMs), e.g., VM**1**, VM**2**, VM**3** and VM**4**, each VM with its own NVMe driver, and a quick emulator (QEMU) for booting the VM. A storage device (e.g., SR-IOV capable SSD) may include four VF apparatuses, namely VF**1**, VF**2**, VF**3** and VF**4** respectively. Each VF apparatus corresponds to a VF queue, which contains I/O commands passed from the corresponding NVMe driver. A controller in the storage device may read the I/O command from each VF queue to read and write data. The reading of the I/O command from the VF queue corresponding to each VF apparatus consumes the corresponding bandwidth resources. Assuming a capacity of 4 T for the entire SSD, each VF may have a capacity of 1 T respectively. Although the capacity of each VF is the same in the example of FIG. **1**, the capacity of each VF may also be uneven. Each VF may be passed-through to the virtual machine via a pass-through method, which reduces the involvement of a virtual machine monitor in I/O operations and improves I/O performance of the virtual machine. However, in such an SR-IOV-based architecture, when load of each VF apparatus is large, there may be competition for bandwidth among the VF apparatuses, and bandwidth resources may be then unevenly distributed.

[0044] FIG. **2** a schematic diagram illustrating an SR-IOV-based I/O virtualization architecture with uneven bandwidth distribution due to I/O competition. As shown in FIG. **2**, with the load of VF**1** and VF**2** being 128k sequential read and 128k sequential write respectively, and VF**3** and VF**4** being 4k random read and 4k random read write respectively, VF**1** and VF**2** may grab more bandwidth resources which may cause VF**3** and VF**4** to occupy very little bandwidth resources, resulting in poorer performance of the virtual system in which VF**3** and VF**4** are located, nonguaranteed quality of service and/or poorer user experience.

[0045] FIG. **3** is a schematic diagram of a device that attempts to achieve balanced bandwidth by adding hardware apparatuses. As shown in FIG. **3**, following hardware modules are added to a device that attempts to achieve balanced bandwidth by adding dedicated hardware apparatus to an existing apparatus: an SSD traffic detection and control unit, which calculates the bandwidth of each controller using data transmission information on bus, and implements speed limit screening and signaling according to the bandwidth ratio set in a traffic control register; an interrupt control unit, which is used to transform the speed limit signal from the traffic control module into an interrupt, and transmit to a CPU core. In this way, the hardware modules may calculate the bandwidth of each controller and, together with the CPU firmware, implement bandwidth constraint function of the virtual apparatus. However, not only does this approach increase the hardware cost, but also this bandwidth constraint approach, although may achieve, by limiting the speed of each VF apparatus, the goal of ensuring that each VF apparatus gets an average bandwidth in case of bandwidth competition without certain VF having poor performance, as to the case of some VFs with large load and some VFs with small load, even if there is unoccupied bandwidth, the VFs with large load cannot use the unoccupied bandwidth due to the speed limit, resulting in a waste of bandwidth resources. The reason is that in existing bandwidth limiting schemes, I/O commands are usually read from VF queues by polling the I/O commands among VF queues according to the number of commands, e.g., the maximum number of commands allowed to be read from each VF queue is set uniformly and if the threshold of the number of commands for each VF queue is exceeded, the I/O command is no longer read from that VF queue. However, such a read approach may result in uneven bandwidth distribution and a waste of bandwidth resources.

[0046] FIG. **4** is a schematic diagram illustrating the bandwidth wastage. As shown in FIG. **4**, VF**2**, VF**3** and VF**4** are all lightly loaded and require only a small amount of bandwidth, while VF**1** is heavily loaded and requires more bandwidth, but because of the bandwidth limit, even if there is unoccupied bandwidth, VF**1** cannot use the unoccupied bandwidth, thus resulting in a waste of bandwidth resources.

[0047] To address this, the present disclosure proposes a conception for dynamic scheduling of bandwidth considering the bandwidth limit and data size of I/O command to avoid the above problem.

[0048] In the following, a storage device and a method performed by the storage device according to the conception of the inventive concepts are described.

[0049] FIG. **5** is a flowchart illustrating a method performed by a storage device according to example embodiments of the present disclosure. According to example embodiments, the storage device may include a memory, and the memory may include a plurality of VF apparatuses. For example, the storage device may be an SSD. Referring to FIG. **5**, in operation S**510**, a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses may be initialized. According to example embodiments, the bandwidth credit of the VF queue corresponding to each VF apparatus may be initialized based on a capacity ratio of each VF apparatus occupying the memory. For example, if a capacity of the storage device is 4 T and a capacity of each VF apparatus is 1 T, the bandwidth credit of each VF queue may be 25% of total bandwidth. Alternatively, it is also possible to define weight values for each VF based on the user's own requirements and then, initialize the bandwidth credits of the VF queues in combination with the weight values. That is, alternatively, the bandwidth credit of the VF queue corresponding to each VF apparatus may be initialized based on the capacity ratio and the weight value (for example, predefined) for each VF apparatus. For example, in the case that a threshold value of bandwidth b (e.g. 25% of the total bandwidth) is set for each VF apparatus according to the capacity ratio, the user is allowed to set a weight w for each VF apparatus according to its own requirements, for example, when a user requires high bandwidth but less capacity for a certain VF apparatus, a value of w greater than **1** may be set to enable that VF

apparatus to obtain a higher bandwidth. In this way, the bandwidth credit of the VF queue corresponding to each VF apparatus may be set in a more personalized way. As an example, the method of setting the weight w may be through the Set Features command or the Vendor Specific command in the NVMe standard, by which the weight of each VF apparatus may be set. The set threshold b and weight w may be stored in the storage device in the form of a table to be used during initialization. For example, the bandwidth credit of the VF queue corresponding to each VF apparatus may be set to be the product of b and w. It should be noted that the way of initializing the bandwidth credit of the VF queue corresponding to each VF apparatus of the inventive concepts is not limited to the above example, but may also be other ways of initializing, for example, directly defining the bandwidth credit of each VF queue by the user.

[0050] After setting the bandwidth credit of the VF queue corresponding to each VF apparatus, in operation S**520**, reading of an I/O command from the VF queue corresponding to each VF apparatus is controlled, based on the bandwidth credit and data size of the I/O command. By considering the bandwidth credit of each VF queue when controlling the reading of the I/O command from the VF queue corresponding to each VF apparatus, uneven bandwidth distribution due to competition for bandwidth resources may be effectively avoided, however, if only the bandwidth credit is considered for command reading control, it may lead to a waste of bandwidth resources. Therefore, the inventive concepts further consider the data size of the I/O command and combines it with the bandwidth credit to jointly control the reading of the I/O command from the VF queue corresponding to each VF apparatus, thereby avoiding both uneven bandwidth distribution and the waste of bandwidth resources, and achieving better dynamic bandwidth scheduling.

[0051] For example, according to example embodiments, operation S**520** may include: polling each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a (for example, preset) threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and in response to the I/O command being read out of the non-empty VF queue, updating the current bandwidth credit of this VF queue by subtracting the data size of the read command from the current bandwidth credit of this VF queue, and in response to the updated current bandwidth credit of this VF queue being no longer greater than the (for example, preset) threshold, stopping reading of the I/O command from that VF queue. For example, the (for example, preset) threshold may be 0, but is not limited thereto. For example, the bandwidth credit of each VF queue may be initialized to credit=w*b, and every time when each command is read out in one VF queue, the data size of the read command is subtracted from the bandwidth credit, and when the current bandwidth credit of one VF queue is less than or equal to 0, no more commands will be read from this VF queue.

[0052] Further, alternatively, the method shown in FIG. **5** may further include: in response to each VF queue being polled once, determining whether current bandwidth credits of all VF queues are all less than or equal to the (for example, preset) threshold; and in response to the current bandwidth credits of all the VF queues all being less than or equal to the (for example, preset) threshold, initializing the bandwidth credit of each VF queue again, otherwise continuing to read the I/O command from the non-empty VF queue with the current bandwidth credit greater than the (for example, preset) threshold. For example, when the credits of all queues are less than or equal to 0, the credits are initialized again and a new round of command reading begins, so that a dynamic balance of bandwidth may be achieved within a cycle (here, the cycle is the time period for the consumption of the bandwidth credits of all queues from the initial value to less than or equal to 0). According to example embodiments, the initializing the bandwidth credit of each VF queue again includes: initializing the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue.

[0053] The method shown in FIG. **5** may further include: setting the current bandwidth credit of a VF queue to the (for example, preset) threshold if this VF queue is empty. Since the I/O command is only read from the non-empty VF queue with the current bandwidth credit greater than the (for example, preset) threshold, after the current bandwidth credit of this VF queue is set to the (for example, preset) threshold, no more commands will be fetched from that queue, and the bandwidth of that queue may be given up for use by other queues in need, thus effectively avoiding a waste of bandwidth resources. For example, when a certain queue which is empty is polled, the credit of that queue will be set to 0, so that the bandwidth given up by that queue may be used by other queues in need, realizing dynamic scheduling of bandwidth resources.

[0054] FIG. **6** is a flowchart illustrating an example of a method performed by a storage device. As shown in FIG. **6**, in operation S**610**, the bandwidth credit credit[i] of each VF queue is initialized to credit[i]=wi*bi, wherein wi is a weight of the ith VF apparatus of the plurality of VF apparatuses, respectively, and bi is a bandwidth limit of the ith VF queue corresponding to the ith VF apparatus. Next, in operation S**620**, it is determined whether credit[i] of each VF queue satisfies credit[i]>0. If credit[i]>0 is satisfied, then in operation S**630**, it is determined whether this VF queue is empty. If credit[i]>0 is not satisfied, proceed to operation S**670**. If this VF queue is empty, then in operation S**640**, the current bandwidth credit of this queue is set to 0, e.g. credit[i]=0. If this VF queue is not empty, then in operation S**650**, the command is read from this VF queue and in operation S**660**, the current bandwidth credit is updated credit[i]=credit[i]−(the data size of the read I/O command). After updating the bandwidth credit, proceed to operation S**670**. In operation S**670**, if credit[i]<=0 for all VF queues are checked. If no, return to perform operation S**620**. If yes, the bandwidth credits for all VF queues are initialized again. For example, in operation S**680**, the bandwidth credits of all VF queues may be initialized again by updating them to credit[i]=credit[i]+wi*bi and then continue with operation S**620**.

[0055] The above method according to example embodiments of the present disclosure enables dynamic bandwidth resource scheduling, which not only achieves balanced bandwidth distribution but also avoids bandwidth wastage. The above method also does require adding may also be provided by a dedicated hardware, for example, an SSD traffic detection control unit, traffic control register, and/or an interrupt control unit.

[0056] FIG. **7** is a schematic diagram of applying the method according to example embodiments of the present disclosure to achieve balanced bandwidth distribution. In the

case of each VF apparatus with heavy load, there will be competition for bandwidth resources, and certain VF apparatuses compete to obtain more resources while certain VF apparatuses become poorly performing and unable to ensure quality of service as a result, however, by applying the above method according to example embodiments of the present disclosure, each VF apparatus has a limit of the bandwidth credit that cannot be exceeded, which ensures that each VF obtains its respective deserved bandwidth in the case of competition for bandwidth. For example, as shown in FIG. **7**, it is assumed that the desired bandwidth of VF**1**, VF**2**, VF**3** and VF**4** is all 25% of the total bandwidth, however, due to competition for bandwidth resources, the actual bandwidth obtained by VF**1**, VF**2**, VF**3** and VF**4** may be 50%, 40%, 5% and 5% respectively. However, after applying the above method according to example embodiments of the present disclosure, ideally, each VF apparatus may obtain approximately 25% of the bandwidth and balanced distribution of bandwidth resources may be substantially achieved.

[0057] FIG. **8** is a schematic diagram of applying a method according to example embodiments of the present disclosure to avoid a waste of bandwidth resources.

[0058] As mentioned above, it may lead to a waste of bandwidth resources that the speed of VF apparatuses with high bandwidth demand is limited even when some of the VF apparatuses are relatively idle. For example, as shown in FIG. **8**, it is supposed that VF**1** is heavily loaded and its desired bandwidth is 85% of the total bandwidth, and VF**2**, VF**3** and VF**4** all have a desired bandwidth of 5%. However, when VF**2**, VF**3** and VF**4** are relatively idle, according to the existing command reading method, even if VF**2**, VF**3** and VF**4** have unoccupied bandwidth, VF**1** cannot use the unoccupied bandwidth, thereby causing a waste of bandwidth. However, according to the above method of example embodiments of the present disclosure, by assigning a bandwidth credit to each VF queue and decreasing the credit according to the data size of each fetched command, when the queue is empty, the credit of the queue is directly set to 0, so that the bandwidth of the idle queue may be given out for use by other queues in need. As shown in FIG. **8**, the desired bandwidth of each VF may ideally be satisfied according to the above method of example embodiments of the present disclosure.

[0059] FIG. **9** is a schematic diagram of the procedure of applying the method according to example embodiments of the present disclosure for dynamic scheduling of bandwidth resources when the load varies.

[0060] In a practical traffic scenario, load on the individual VFs is dynamically changing, and the load may be small at first, and there may be a sudden increase in load, for example, as shown in FIG. **9**, initially only the load on VF**1** is large and the load on VF**2**, VF**3**, and VF**4** are all small, at this moment, according to the above method of example embodiments of the present disclosure, VF**1** may use the bandwidth given up by idle VFs (e.g., VF**2**, VF**3** and VF**4**). Then, the load on VF**2** suddenly becomes large due to traffic demand, and according to the above method of example embodiments of the present disclosure, all available bandwidth may be dynamically allocated to the two VFs (e.g., VF**1** and VF**2**) in need in a balanced manner. Then, as the traffic changes, a larger load is gradually generated on VF**3** and VF**4**, and finally, in the case that all VFs need to grab the bandwidth, the bandwidth may be allocated to the four VFs in a balanced manner according to the above method of example embodiments of the present disclosure.

[0061] Hereinafter, the storage device according to example embodiments of the present disclosure is described with reference to FIG. **10** and FIG. **11**. FIG. **10** illustrates a block diagram of a storage device according to example embodiments of the present disclosure.

[0062] Referring to FIG. **10**, the storage device **1000***a* includes a memory **1010** and a controller **1020**. The memory **1010** may include a plurality of virtual function (VF) devices. The controller **1020** may be configured to: initialize a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and control reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and data size of the I/O command.

[0063] For example, according to example embodiments, the controller **1020** may be configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on a capacity ratio of each VF apparatus occupying the memory **1010**. Alternatively, the controller **1020** may be configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on the capacity ratio and a weight value (for example, predefined) for each VF apparatus.

[0064] Further, according to example embodiments, the controller **1020** may be configured to: poll each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a (for example, preset) threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and in response to the I/O command being read out of the non-empty VF queue, update the current bandwidth credit of this VF queue by subtracting the data size of the read command from the current bandwidth credit of this VF queue, and in response to the updated current bandwidth credit of this VF queue being no longer greater than the (for example, preset) threshold, stop reading of the I/O command from this VF queue.

[0065] Alternatively, according to example embodiments, the controller **1020** may further be configured to: in response to each VF queue being polled once, determine whether current bandwidth credits of all VF queues are all less than or equal to the (for example, preset) threshold; in response to the current bandwidth credits of all the VF queues all being less than or equal to the (for example, preset) threshold, initialize the bandwidth credit of each VF queue again, otherwise continue to read the I/O command from the non-empty VF queue with the current bandwidth credit greater than the (for example, preset) threshold. For example, when the current bandwidth credits of all the VF queues are less than or equal to the (for example, preset) threshold, the controller **1020** initializes the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue. Alternatively, according to example embodiments, the controller **1020** may also be configured to set the current bandwidth credit of a VF queue to the (for example, preset) threshold if this VF queue is empty.

[0066] FIG. **11** illustrates an example schematic diagram of a storage device according to example embodiments of the present disclosure. As shown in FIG. **11**, the storage device **1000***a* may be an SSD, e.g., an SR-IOV capable SSD. For example, the memory **1010** included in storage device

1000*a* may be an array of NAND chips. A plurality of VF apparatuses may be included in the array of NAND chips. The controller **1020** included in the storage device **1000***a* may be a primary controller in the SSD. The primary controller may include a dynamic bandwidth scheduler module which may be configured to perform the operations mentioned above which are performed by the controller **1010**. For example, this dynamic bandwidth scheduler module may initialize the bandwidth credit of the VF queue corresponding to each VF apparatus of the plurality of VF apparatuses, for example, initializing the bandwidth credit of each VF queue to the weight*the bandwidth limit. This dynamic bandwidth scheduler module may then control the reading of I/O command from the VF queues corresponding to each VF apparatus based on the bandwidth credit and the data size of the I/O commands, thereby enabling dynamic bandwidth scheduling.

[0067] The storage device according to example embodiments of the present disclosure may achieve dynamic bandwidth resource scheduling, which not only achieves a balanced bandwidth distribution, but also avoids bandwidth wastage.

[0068] FIG. **12** is a diagram of a system **1000***b* to which a storage device is applied, according to example embodiments of the present disclosure.

[0069] The system **1000***b* of FIG. **12** may basically be a mobile system, such as a portable communication terminal (e.g., a mobile phone), a smartphone, a tablet personal computer (PC), a wearable device, a healthcare device, or an Internet of things (IOT) device. However, the system **1000***b* of FIG. **12** is not necessarily limited to a mobile system and may be a PC, a laptop computer, a server, a media player, or an automotive device (e.g., a navigation device).

[0070] Referring to FIG. **12**, the system **1000***b* may include a main processor **1100**, memories (e.g., **1200***a* and **1200***b*), and storage devices (e.g., **1300***a* and **1300***b*). The storage devices may be configured to perform the data access method described above. The system **1000***b* may include at least one of an image capturing device **1410**, a user input device **1420**, a sensor **1430**, a communication device **1440**, a display **1450**, a speaker **1460**, a power supplying device **1470**, and a connecting interface **1480**.

[0071] The main processor **1100** may control all operations of the system **1000***b*, for example, operations of other components included in the system **1000***b*. The main processor **1100** may be implemented as a general-purpose processor, a dedicated processor, or an application processor.

[0072] The main processor **1100** may include at least one CPU core **1110** and further include a controller **1120** configured to control the memories **1200***a* and **1200***b* and/or the storage devices **1300***a* and **1300***b*. In some example embodiments, the main processor **1100** may further include an accelerator **1130**, which is a dedicated circuit for a high-speed data operation, such as an artificial intelligence (AI) data operation. The accelerator **1130** may include a graphics processing unit (GPU), a neural processing unit (NPU) and/or a data processing unit (DPU) and be implemented as a chip that is physically separate from the other components of the main processor **1100**.

[0073] The accelerator **1130** may, for example, have a structure that is trainable, e.g., with training data, such as an artificial neural network, a decision tree, a support vector machine, a Bayesian network, a genetic algorithm, and/or the like. Non-limiting examples of the trainable structure may include a convolution neural network (CNN), a generative adversarial network (GAN), an artificial neural network (ANN), a region based convolution neural network (R-CNN), a region proposal network (RPN), a recurrent neural network (RNN), a stacking-based deep neural network (S-DNN), a state-space dynamic neural network (S-SDNN), a deconvolution network, a deep belief network (DBN), a restricted Boltzmann machine (RBM), a fully convolutional network, a long short-term memory (LSTM) network, a classification network, and/or the like.

[0074] The memories **1200***a* and **1200***b* may be used as main memory devices of the system **1000***b*. Although each of the memories **1200***a* and **1200***b* may include a volatile memory, such as static random access memory (SRAM) and/or dynamic RAM (DRAM), each of the memories **1200***a* and **1200***b* may include non-volatile memory, such as a flash memory, phase-change RAM (PRAM) and/or resistive RAM (RRAM). The memories **1200***a* and **1200***b* may be implemented in the same package as the main processor **1100**.

[0075] The storage devices **1300***a* and **1300***b* may serve as non-volatile storage devices configured to store data regardless of whether power is supplied thereto, and have larger storage capacity than the memories **1200***a* and **1200***b*. The storage devices **1300***a* and **1300***b* may respectively include storage controllers (STRG CTRL) **1310***a* and **1310***b* and NVMs (Non-Volatile Memories) **1320***a* and **1320***b* configured to store data via the control of the storage controllers **1310***a* and **1310***b*. Although the NVMs **1320***a* and **1320***b* may include flash memories having a two-dimensional (2D) structure or a three-dimensional (3D) V-NAND structure, the NVMs **1320***a* and **1320***b* may include other types of NVMs, such as PRAM and/or RRAM.

[0076] The storage devices **1300***a* and **1300***b* may be physically separated from the main processor **1100** and included in the system **1000***b* or implemented in the same package as the main processor **1100**. The storage devices **1300***a* and **1300***b* may have types of solid-state devices (SSDs) or memory cards and be removably combined with other components of the system **100** through an interface, such as the connecting interface **1480** that will be described below. The storage devices **1300***a* and **1300***b* may be devices to which a standard protocol, such as a universal flash storage (UFS), an embedded multi-media card (eMMC), or a non-volatile memory express (NVMe), is applied, without being limited thereto.

[0077] The image capturing device **1410** may capture still images or moving images. The image capturing device **1410** may include a camera, a camcorder, and/or a webcam.

[0078] The user input device **1420** may receive various types of data input by a user of the system **1000***b* and include a touch pad, a keypad, a keyboard, a mouse, and/or a microphone.

[0079] The sensor **1430** may detect various types of physical quantities, which may be obtained from the outside of the system **1000***b*, and convert the detected physical quantities into electric signals. The sensor **1430** may include a temperature sensor, a pressure sensor, an illuminance sensor, a position sensor, an acceleration sensor, a biosensor, and/or a gyroscope sensor.

[0080] The communication device **1440** may transmit and receive signals between other devices outside the system

1000*b* according to various communication protocols. The communication device **1440** may include an antenna, a transceiver, and/or a modem.

[0081] The display **1450** and the speaker **1460** may serve as output devices configured to respectively output visual information and auditory information to the user of the system **1000***b*.

[0082] The power supplying device **1470** may appropriately convert power supplied from a battery (not shown) embedded in the system **1000***b* and/or an external power source, and supply the converted power to each of components of the system **1000***b*.

[0083] The connecting interface **1480** may provide connection between the system **1000***b* and an external device, which is connected to the system **1000***b* and capable of transmitting and receiving data to and from the system **1000***b*. The connecting interface **1480** may be implemented by using various interface schemes, such as advanced technology attachment (ATA), serial ATA (SATA), external SATA (e-SATA), small computer small interface (SCSI), serial attached SCSI (SAS), peripheral component interconnection (PCI), PCI express (PCIe), NVMe, IEEE 1394, a universal serial bus (USB) interface, a secure digital (SD) card interface, a multi-media card (MMC) interface, an eMMC interface, a UFS interface, an embedded UFS (eUFS) interface, and a compact flash (CF) card interface.

[0084] The storage device (e.g., **1300***a* or **1300***b*) may be a solid state drive (SSD). According to example embodiments of the present disclosure, a system (e.g., **1000***b*), to which a storage device is applied, is provided, including: a main processor (e.g., **1100**); a memory (e.g., **1200***a* and **1200***b*); and the storage device (e.g., **1300***a* and **1300***b*), wherein the storage device is configured to perform the method according to example embodiments of the present disclosure as described above.

[0085] FIG. **13** is a block diagram of a host storage system **10** according to example embodiments.

[0086] The host storage system **10** may include a host **100** and a storage device **200**. The storage device **200** may be configured to perform the data access method described above. Further, the storage device **200** may include a storage controller **210** and an NVM **220**. According to example embodiments, the host **100** may include a host controller **110** and a host memory **120**. The host memory **120** may serve as a buffer memory configured to temporarily store data to be transmitted to the storage device **200** or data received from the storage device **200**.

[0087] The storage device **200** may include storage media configured to store data in response to requests from the host **100**. As an example, the storage device **200** may include at least one of an SSD, an embedded memory, and a removable external memory. When the storage device **200** is an SSD, the storage device **200** may be a device that conforms to an NVMe standard. When the storage device **200** is an embedded memory or an external memory, the storage device **200** may be a device that conforms to a UFS standard or an eMMC standard. Each of the host **100** and the storage device **200** may generate a packet according to an adopted standard protocol and transmit the packet.

[0088] When the NVM **220** of the storage device **200** includes a flash memory, the flash memory may include a 2D NAND memory array or a 3D (or vertical) NAND (VNAND) memory array. As another example, the storage device **200** may include various other kinds of NVMs. For example, the storage device **200** may include magnetic RAM (MRAM), spin-transfer torque MRAM, conductive bridging RAM (CBRAM), ferroelectric RAM (FRAM), PRAM, RRAM, and various other kinds of memories.

[0089] According to example embodiments, the host controller **110** and the host memory **120** may be implemented as separate semiconductor chips. Alternatively, in some example embodiments, the host controller **110** and the host memory **120** may be integrated in the same semiconductor chip. As an example, the host controller **110** may be any one of a plurality of modules included in an application processor (AP). The AP may be implemented as a System on Chip (SoC). Further, the host memory **120** may be an embedded memory included in the AP or an NVM or memory module located outside the AP.

[0090] The host controller **110** may manage an operation of storing data (e.g., write data) of a buffer region of the host memory **120** in the NVM **220** or an operation of storing data (e.g., read data) of the NVM **220** in the buffer region.

[0091] The storage controller **210** may include a host interface **211**, a memory interface **212**, and a CPU **213**. Further, the storage controllers **210** may further include a flash translation layer (FTL) **214**, a packet manager **215**, a buffer memory **216**, an error correction code (ECC) engine **217**, and an advanced encryption standard (AES) engine **218**. The storage controllers **210** may further include a working memory (not shown) in which the FTL **214** is loaded. The CPU **213** may execute the FTL **214** to control data write and read operations on the NVM **220**.

[0092] The host interface **211** may transmit and receive packets to and from the host **100**. A packet transmitted from the host **100** to the host interface **211** may include a command or data to be written to the NVM **220**. A packet transmitted from the host interface **211** to the host **100** may include a response to the command or data read from the NVM **220**. The memory interface **212** may transmit data to be written to the NVM **220** to the NVM **220** or receive data read from the NVM **220**. The memory interface **212** may be configured to comply with a standard protocol, such as Toggle or open NAND flash interface (ONFI).

[0093] The FTL **214** may perform various functions, such as an address mapping operation, a wear-leveling operation, and a garbage collection operation. The address mapping operation may be an operation of converting a logical address received from the host **100** into a physical address used to actually store data in the NVM **220**. The wear-leveling operation may be a technique for reducing or preventing excessive deterioration of a specific block by allowing blocks of the NVM **220** to be more uniformly used. As an example, the wear-leveling operation may be implemented using a firmware technique that balances erase counts of physical blocks. The garbage collection operation may be a technique for ensuring usable capacity in the NVM **220** by erasing an existing block after copying valid data of the existing block to a new block.

[0094] The packet manager **215** may generate a packet according to a protocol of an interface, which consents to the host **100**, or parse various types of information from the packet received from the host **100**. The buffer memory **216** may temporarily store data to be written to the NVM **220** or data to be read from the NVM **220**. Although the buffer memory **216** may be a component included in the storage controllers **210**, the buffer memory **216** may be outside the storage controllers **210**.

[0095] The ECC engine **217** may perform error detection and correction operations on read data read from the NVM **220**. For example, the ECC engine **217** may generate parity bits for write data to be written to the NVM **220**, and the generated parity bits may be stored in the NVM **220** together with write data. During the reading of data from the NVM **220**, the ECC engine **217** may correct an error in the read data by using the parity bits read from the NVM **220** along with the read data, and output error-corrected read data.

[0096] The AES engine **218** may perform at least one of an encryption operation and a decryption operation on data input to the storage controllers **210** by using a symmetric-key algorithm.

[0097] The storage device **200** may be a solid state drive (SSD). According to example embodiments of the present disclosure, a host storage system (e.g., **10**) is provided, including: a host (e.g., **100**); and a storage device (**200**), wherein the storage device is configured to perform the method according to example embodiments of the present disclosure as described above.

[0098] FIG. **14** is a diagram of a data center **3000** to which a memory device is applied, according to example embodiments of the present disclosure.

Platform Portion—Server (Application/Storage)

[0099] Referring to FIG. **14**, the data center **3000** may be a facility that collects various types of pieces of data and provides services and be referred to as a data storage center. The data center **3000** may be a system for operating a search engine and a database, and may be a computing system used by companies, such as banks, or government agencies. The data center **3000** may include application servers **3100** to **3100***n* and storage servers **3200** to **3200***m*. The number of application servers **3100** to **3100***n* and the number of storage servers **3200** to **3200***m* may be variously selected according to example embodiments. The number of application servers **3100** to **3100***n* may be different from the number of storage servers **3200** to **3200***m*.

[0100] The application server **3100** or the storage server **3200** may include at least one of processors **3110** and **3210** and memories **3120** and **3220**. The storage server **3200** will now be described as an example. The processor **3210** may control all operations of the storage server **3200**, access the memory **3220**, and execute instructions and/or data loaded in the memory **3220**. The memory **3220** may be a double-data-rate synchronous DRAM (DDR SDRAM), a high-bandwidth memory (HBM), a hybrid memory cube (HMC), a dual in-line memory module (DIMM), Optane DIMM, and/or a non-volatile DIMM (NVMDIMM). In some example embodiments, the numbers of processors **3210** and memories **3220** included in the storage server **3200** may be variously selected. In example embodiments, the processor **3210** and the memory **3220** may provide a processor-memory pair. In example embodiments, the number of processors **3210** may be different from the number of memories **3220**. The processor **3210** may include a single-core processor or a multi-core processor. The above description of the storage server **3200** may be similarly applied to the application server **3100**. In some example embodiments, the application server **3100** may not include a storage device **3150**. The storage server **3200** may include at least one storage device **3250**. The number of storage devices **3250** included in the storage server **3200** may be variously selected according to example embodiments.

Platform Portion—Network

[0101] The application servers **3100** to **3100***n* may communicate with the storage servers **3200** to **3200***m* through a network **3300**. The network **3300** may be implemented by using a fiber channel (FC) or Ethernet. In some example embodiments, the FC may be a medium used for relatively high-speed data transmission and use an optical switch with higher performance and/or higher availability. The storage servers **3200** to **3200***m* may be provided as file storages, block storages, or object storages according to an access method of the network **3300**.

[0102] In example embodiments, the network **3300** may be a storage-dedicated network, such as a storage area network (SAN). For example, the SAN may be an FC-SAN, which uses an FC network and is implemented according to an FC protocol (FCP). As another example, the SAN may be an Internet protocol (IP)-SAN, which uses a transmission control protocol (TCP)/IP network and is implemented according to a SCSI over TCP/IP or Internet SCSI (iSCSI) protocol. In other example embodiments, the network **3300** may be a general network, such as a TCP/IP network. For example, the network **3300** may be implemented according to a protocol, such as FC over Ethernet (FCOE), network attached storage (NAS), and NVMe over Fabrics (NVMe-oF).

[0103] Hereinafter, the application server **3100** and the storage server **3200** will mainly be described. A description of the application server **3100** may be applied to another application server **3100***n*, and a description of the storage server **3200** may be applied to another storage server **3200***m*.

[0104] The application server **3100** may store data, which is requested by a user or a client to be stored, in one of the storage servers **3200** to **3200***m* through the network **3300**. Also, the application server **3100** may obtain data, which is requested by the user or the client to be read, from one of the storage servers **3200** to **3200***m* through the network **3300**. For example, the application server **3100** may be implemented as a web server or a database management system (DBMS).

[0105] The application server **3100** may access a memory **3120***n* or a storage device **3150***n*, which is included in another application server **3100***n*, through the network **3300**. Alternatively, the application server **3100** may access memories **3220** to **3220***m* or storage devices **3250** to **3250***m*, which are included in the storage servers **3200** to **3200***m*, through the network **3300**. Thus, the application server **3100** may perform various operations on data stored in application servers **3100** to **3100***n* and/or the storage servers **3200** to **3200***m*. For example, the application server **3100** may execute an instruction for moving or copying data between the application servers **3100** to **3100***n* and/or the storage servers **3200** to **3200***m*. In some example embodiments, the data may be moved from the storage devices **3250** to **3250***m* of the storage servers **3200** to **3200***m* to the memories **3120** to **3120***n* of the application servers **3100** to **3100***n* directly or through the memories **3220** to **3220***m* of the storage servers **3200** to **3200***m*. The data moved through the network **3300** may be data encrypted for security or privacy.

Organic Relationship—Interface Structure/Type

[0106] The storage server **3200** will now be described as an example. An interface **3254** may provide physical connection between a processor **3210** and a controller **3251** and

a physical connection between a network interface card (NIC) **3240** and the controller **3251**. For example, the interface **3254** may be implemented using a direct attached storage (DAS) scheme in which the storage device **3250** is directly connected with a dedicated cable. For example, the interface **3254** may be implemented by using various interface schemes, such as ATA, SATA, e-SATA, an SCSI, SAS, PCI, PCIe, NVMe, IEEE 1394, a USB interface, an SD card interface, an MMC interface, an eMMC interface, a UFS interface, an eUFS interface, and/or a CF card interface.

[0107] The storage server **3200** may further include a switch **3230** and the NIC (Network InterConnect) **3240**. The switch **3230** may selectively connect the processor **3210** to the storage device **3250** or selectively connect the NIC **3240** to the storage device **3250** via the control of the processor **3210**.

[0108] In example embodiments, the NIC **3240** may include a network interface card and a network adaptor. The NIC **3240** may be connected to the network **3300** by a wired interface, a wireless interface, a Bluetooth interface, or an optical interface. The NIC **3240** may include an internal memory, a digital signal processor (DSP), and a host bus interface and be connected to the processor **3210** and/or the switch **3230** through the host bus interface. The host bus interface may be implemented as one of the above-described examples of the interface **3254**. In example embodiments, the NIC **3240** may be integrated with at least one of the processor **3210**, the switch **3230**, and the storage device **3250**.

Organic Relationship—Interface Operation

[0109] In the storage servers **3200** to **3200***m* or the application servers **3100** to **3100***n*, a processor may transmit a command to storage devices **3150** to **3150***n* and **3250** to **3250***m* or the memories **3120** to **3120***n* and **3220** to **3220***m* and program or read data. In some example embodiments, the data may be data of which an error is corrected by an ECC engine. The data may be data on which a data bus inversion (DBI) operation or a data masking (DM) operation is performed, and may include cyclic redundancy code (CRC) information. The data may be data encrypted for security or privacy.

[0110] Storage devices **3150** to **3150***n* and **3250** to **3250***m* may transmit a control signal and a command/address signal to NAND flash memory devices **3252** to **3252***m* in response to a read command received from the processor. Thus, in response to data being read from the NAND flash memory devices **3252** to **3252***m*, a read enable (RE) signal may be input as a data output control signal, and thus, the data may be output to a DQ bus. A data strobe signal DQS may be generated using the RE signal. The command and the address signal may be latched in a page buffer depending on a rising edge or falling edge of a write enable (WE) signal.

Product Portion—SSD Basic Operation

[0111] The controller **3251** may control all operations of the storage device **3250**. In example embodiments, the controller **3251** may include SRAM. The controller **3251** may write data to the NAND flash memory device **3252** in response to a write command or read data from the NAND flash memory device **3252** in response to a read command. For example, the write command and/or the read command may be provided from the processor **3210** of the storage server **3200**, the processor **3210***m* of another storage server **3200***m*, or the processors **3110** and **3110***n* of the application servers **3100** and **3100***n*. DRAM **3253** may temporarily store (or buffer) data to be written to the NAND flash memory device **3252** or data read from the NAND flash memory device **3252**. Also, the DRAM **3253** may store metadata. Here, the metadata may be user data or data generated by the controller **3251** to manage the NAND flash memory device **3252**. The storage device **3250** may include a secure element (SE) for security or privacy.

[0112] The storage device **200** may be an SSD. According to example embodiments of the present disclosure, a data center system (e.g., **3000**) is provided, including: a plurality of application servers (**3100** to **3100***n*); and a plurality of storage servers (e.g., **3200** to **3200***m*), wherein each storage server includes a storage device **200**, wherein the storage device **200** is configured to perform the method according to example embodiments of the present disclosure as described above.

[0113] According to example embodiments of the present disclosure, a computer readable storage medium having a computer program stored thereon is provided, wherein the method according to example embodiments of the present disclosure described above is implemented when the computer program is executed by a processor.

[0114] According to example embodiments of the present disclosure, an electronic apparatus is provided, including: a processor; a memory storing a computer program, wherein the computer program when executed by the processor implements the method according to example embodiments of the present disclosure described above.

[0115] According to example embodiments of the present disclosure, a computer-readable storage medium storing instructions may also be provided, the instructions, when executed by at least one processor, cause the at least one processor to perform the method according to example embodiments of the present disclosure described above. Examples of computer-readable storage media herein include: read-only memory (ROM), random access programmable read-only memory (PROM), electrically erasable programmable read-only memory (EEPROM), random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), flash memory, non-volatile memory, CD-ROM, CD-R, CD+R, CD-RW, CD+RW, DVD-ROM, DVD-R, DVD+R, DVD-RW, DVD+RW, DVD-RAM, BD-ROM, BD-R, BD-R LTH, BD-RE, Blu-ray or optical disk memory, hard disk drive (HDD), solid state drive (SSD), card-based memory (such as, multimedia cards, Secure Digital (SD) cards and/or Extreme Digital (XD) cards), magnetic tapes, floppy disks, magneto-optical data storage devices, optical data storage devices, hard disks, solid state disks, and/or any other device, where the other device is configured to store the computer programs and any associated data, data files, and/or data structures in a non-transitory manner and to provide the computer programs and any associated data, data files, and/or data structures to a processor or computer, so that the processor or computer may execute the computer program. The computer program in the computer readable storage medium may run in an environment deployed in a computer device such as a terminal, client, host, agent, server, etc., and furthermore, in one example, the computer program and any associated data, data files and data structures are distributed on a networked computer system such

that the computer program and any associated data, data files and data structures are stored, accessed, and/or executed in a distributed manner by one or more processors or computers.

[0116] One or more elements described above may be implemented using processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc. The processing circuitry may include a memory such as a volatile memory device (e.g., SRAM, DRAM, SDRAM, etc.) and/or a non-volatile memory (e.g., flash memory device, phase-change memory, ferroelectric memory device, etc.).

[0117] Other example embodiments of the present disclosure will readily come to the mind of those skilled in the art upon consideration of the specification and practice of the present disclosure disclosed herein. This application is intended to cover any variations, uses, or adaptations of the present disclosure that follow the general principles of the present disclosure and include commonly known or customary technical means in the art that are not disclosed herein. The specification and example embodiments are considered examples only, and the true scope and spirit of the present disclosure is indicated by the following claims.

1. A storage device comprising:

a memory comprising a plurality of virtual function (VF) apparatuses;

a controller configured to:

initialize a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and

control reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and data size of the I/O command.

2. The storage device of claim **1**, wherein the controller is configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on a capacity ratio of each VF apparatus occupying the memory.

3. The storage device of claim **2**, wherein the controller is configured to initialize the bandwidth credit of the VF queue corresponding to each VF apparatus based on the capacity ratio and a weight value for each VF apparatus.

4. The storage device of claim **1**, wherein the controller is configured to:

poll each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and

in response to the I/O command being read out of the non-empty VF queue, update the current bandwidth credit of the VF queue by subtracting the data size of the read command from the current bandwidth credit of the VF queue, and in response to the updated current bandwidth credit of the VF queue being no longer greater than the threshold, stop reading of the I/O command from the VF queue.

5. The storage device of claim **4**, wherein the controller is further configured to:

in response to each VF queue being polled once, determine whether current bandwidth credits of all VF queues are all less than or equal to the threshold;

in response to the current bandwidth credits of all the VF queues all being less than or equal to the threshold, initialize the bandwidth credit of each VF queue again, otherwise continue to read the I/O command from the non-empty VF queue with the current bandwidth credit greater than the threshold.

6. The storage device of claim **5**, wherein the controller is further configured to:

set the current bandwidth credit of a VF queue to the threshold if the VF queue is empty.

7. The storage device of claim **5**, wherein in response to the current bandwidth credits of all the VF queues being less than or equal to the threshold, the controller is configured to initialize the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue.

8. A method performed by a storage device, wherein the storage device comprises a memory, the memory comprising a plurality of virtual function (VF) apparatuses, the method comprising:

initializing a bandwidth credit of a VF queue corresponding to each VF apparatus of the plurality of VF apparatuses; and

controlling reading of an I/O command from the VF queue corresponding to each VF apparatus, based on the bandwidth credit and the data size of the I/O command.

9. The method of claim **8**, wherein the initializing the bandwidth credit of the VF queue corresponding to each VF apparatus of the plurality of VF apparatuses comprises:

initializing the bandwidth credit of the VF queue corresponding to each VF apparatus based on a capacity ratio of each VF apparatus occupying the memory.

10. The method of claim **9**, wherein the initializing the bandwidth credit of the VF queues corresponding to each VF apparatus based on the capacity ratio of each VF apparatus occupying the memory comprises: initializing the bandwidth credit of the VF queue corresponding to each VF apparatus based on the capacity ratio and a weight value for each VF apparatus.

11. The method of claim **8**, wherein the controlling the reading of the I/O command from the VF queue corresponding to each VF apparatus based on the bandwidth credit and the data size of the I/O commands comprises:

polling each VF queue to read the I/O command from a non-empty VF queue with a current bandwidth credit greater than a threshold based on initializing the bandwidth credit of the VF queue corresponding to each VF apparatus; and

in response to the I/O command being read out of the non-empty VF queue, updating the current bandwidth credit of the VF queue by subtracting the data size of the read command from the current bandwidth credit of the VF queue, and in response to the updated current bandwidth credit of the VF queue being no longer greater than the threshold, stopping reading of the I/O command from the VF queue.

12. The method of claim 11, further comprising:

in response to each VF queue being polled once, determine whether current bandwidth credits of all VF queues are all less than or equal to the threshold;

in response to the current bandwidth credits of all the VF queues all being less than or equal to the threshold, initializing the bandwidth credit of each VF queue again, otherwise continuing to read I/O commands from non-empty VF queues with the current bandwidth credit greater than the threshold.

13. The method of claim 12, further comprising:

setting the current bandwidth credit of a VF queue to the threshold if the VF queue is empty.

14. The method of claim 12, wherein the initializing the bandwidth credit of each VF queue again comprises: initializing the bandwidth credit of each VF queue again by adding the current bandwidth credit of each VF queue to a first initialized bandwidth credit of each VF queue.

15. A storage system comprising:

a main processor;

a main memory; and

a storage device,

wherein the storage device is configured to perform the method according to claim 8.

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

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